Supermarket Sales Analysis & Business Insights Report

1. Project Overview

This project involves conducting a detailed Exploratory Data Analysis (EDA) of sales transactions from a regional supermarket network in New York, Los Angeles, and Chicago.

The objective is to analyze historical sales data to understand customer purchasing behaviors, assess the performance of various product categories, and identify opportunities to boost profitability and improve operational efficiency.

2. Data Description

The dataset was obtained from the supermarket's internal sales system, encompassing various transactions across its branches.

Data Dimensions: 8 columns and 254 individual transaction records.

Variable Definitions:

Sale_id: Unique identifier for each transaction.

Branch: Brand of the supermarket.

City: Location of the branch.

Customer_type: Segmentation of customers (Member vs. Normal).

Product_name: The specific item purchased.

Product_category: Broad classification of the product.

Quantity: The number of units sold.

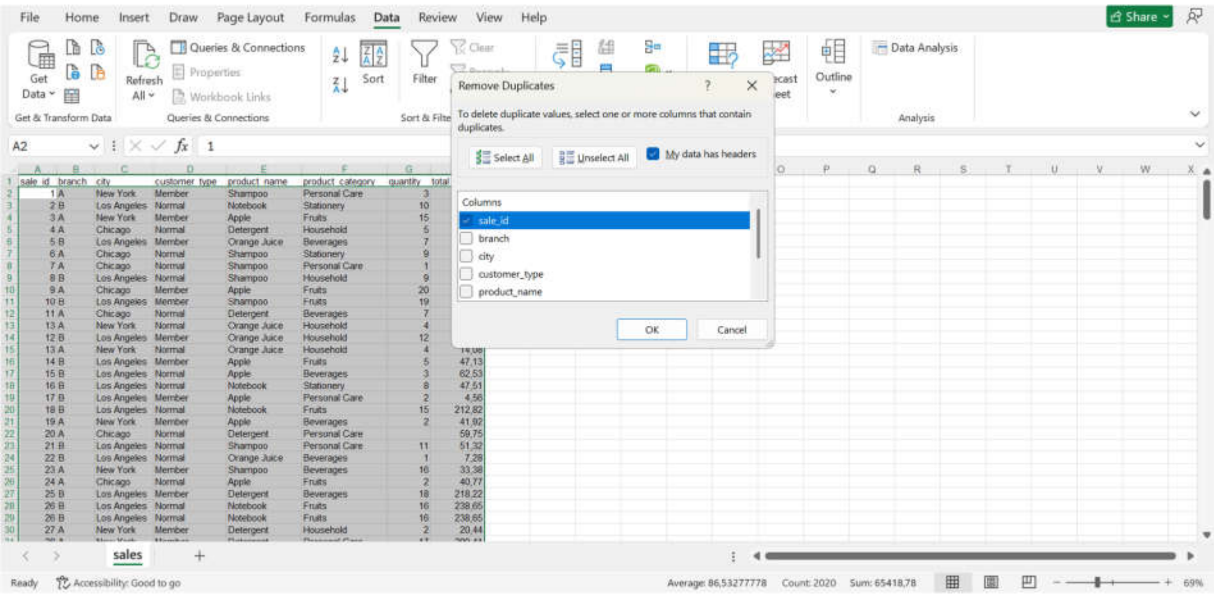
Total_price: Final transaction amount in USD.

3. Data Cleaning & Preparation

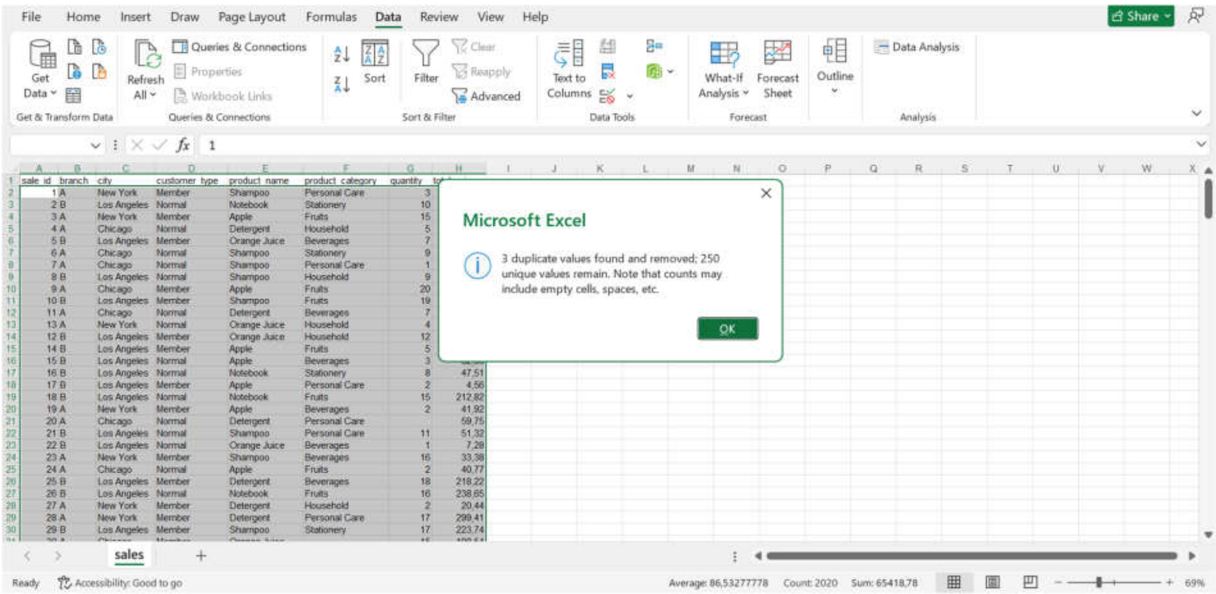
To ensure high-quality analysis, a rigorous data cleaning process was performed using Excel:

- *Removing duplicate records:*

I used the 'Remove duplicate records' feature with Sale_id as the primary key.



Three duplicate records were identified and removed to avoid double-calculating sales.



- *Data consistency and standardization:*

I discovered significant anomalies in data entry in the Product Category column. Several items were severely misclassified (e.g., "Apples" were incorrectly categorized under "Personal Care" or "Stationery," and "Notebooks" were classified as "Beverages").

product_name	product_category	product_name	product_category
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Fruits
Apple	Beverages	Notebook	Fruits
Apple	Personal Care	Notebook	Stationery
Apple	Beverages	Notebook	Beverages
Apple	Fruits	Notebook	Beverages
Apple	Personal Care	Notebook	Household
Apple	Fruits	Notebook	Fruits
Apple	Household	Notebook	Household
Apple	Stationery	Notebook	Fruits
Apple	Personal Care	Notebook	Stationery

I systematically checked the Product Name column and reassigned any misclassified transactions to their correct, standardized categories. The following strict classification rules were applied:

Apple ➔ Classified only in the Fruits category.

Detergent ➔ Classified only in the Household category.

Notebook ➔ Classified only in the Stationery category.

Orange Juice ➔ Classified only in the Beverages category.

Shampoo ➔ Classified only in the Personal Care category.

product_name	product_category	product_name	product_category
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery
Apple	Fruits	Notebook	Stationery

- *Handling Missing Values:*

Missing data points were handled using separate methods, tailored to the data to preserve both statistical validity and underlying business logic:

Fill in Categorical Value (Customer Type): Several records were missing customer segment data.

29	28	A	New York	Member
30	29	B	Los Angeles	Member
31	30	A	Chicago	Member
32	31	A	Chicago	
33	32	A	New York	Member
34	33	B	Los Angeles	Normal
35	34	B	Los Angeles	Normal
36	35	A	New York	
37	36	A	New York	Member
38	37	B	Los Angeles	Member
39	38	B	Los Angeles	Member
40	40	A	New York	Normal
41	39	A	New York	Member
42	41	A	New York	Member
43	42	A	New York	Normal
44	43	A	New York	Normal
45	44	A	Chicago	
46	45	B	Los Angeles	Member
47	46	A	New York	Member
48	47	A	New York	Normal

sales

To maintain dataset balance and avoid unnecessary row deletions, I used "mode imputation" by replacing blank cells with the mode value ("Member").

30	29	B	Los Angeles	Member
31	30	A	Chicago	Member
32	31	A	Chicago	Member
33	32	A	New York	Member
34	33	B	Los Angeles	Normal
35	34	B	Los Angeles	Normal
36	35	A	New York	Member
37	36	A	New York	Member
38	37	B	Los Angeles	Member
39	38	B	Los Angeles	Member
40	40	A	New York	Normal
41	39	A	New York	Member
42	41	A	New York	Member
43	42	A	New York	Normal
44	43	A	New York	Normal
45	44	A	Chicago	Member
46	45	B	Los Angeles	Member
47	46	A	New York	Member
48	47	A	New York	Normal
49	48	B	Los Angeles	Normal

sales

Filling in Numerical Value (Quantity): Missing sales volume data was identified in three specific transactions (Sale_id 20, 44, 61), all related to the product "Detergent".

	A	B	C	D	E	F	G	H
1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
21	20	A	Chicago	Normal	Detergent	Household		59,75
45	44	A	Chicago	Member	Detergent	Household		43,08
62	61	A	New York	Member	Detergent	Household		20,09

I noticed that the unit price varies between different orders due to various factors (such as branch-specific pricing or applied discounts), so the unit price alone cannot be used to calculate the quantity of detergent sold for those orders. An average unit price needs to be calculated as a basis. I calculated the total detergent sold and its total value, excluding the three missing records, using the "Sum" function. Then I divided the sum of the total price by the total number of detergents.

Detergent	Household	18	229,58	
Detergent	Household	18	272,91	
Detergent	Household	12	123,65	
Detergent	Household	12	68,82	
Detergent	Household	15	277,99	
Detergent	Household	8	26,88	
Detergent	Household	10	58,85	
Detergent	Household	13	115,45	
Detergent	Household	8	93,3	
Detergent	Household	10	176,55	
Detergent	Household	18	162,94	
Detergent	Household	17	227,38	
Detergent	Household	9	192,21	
Detergent	Household	13	66,77	
Detergent	Household	8	154,94	
Detergent	Household	3	28,28	
Detergent	Household	19	195,37	
		Total quantity	Sum of Total price	Average Unit price
		2575	**30784,06**	**11,95497476**

Finally, to get the quantity of the missing orders, I divided the total price of each order by the average unit price, using the round function to round if the number is odd.

G62 =ROUND([@[total_price]]/11,95;0)

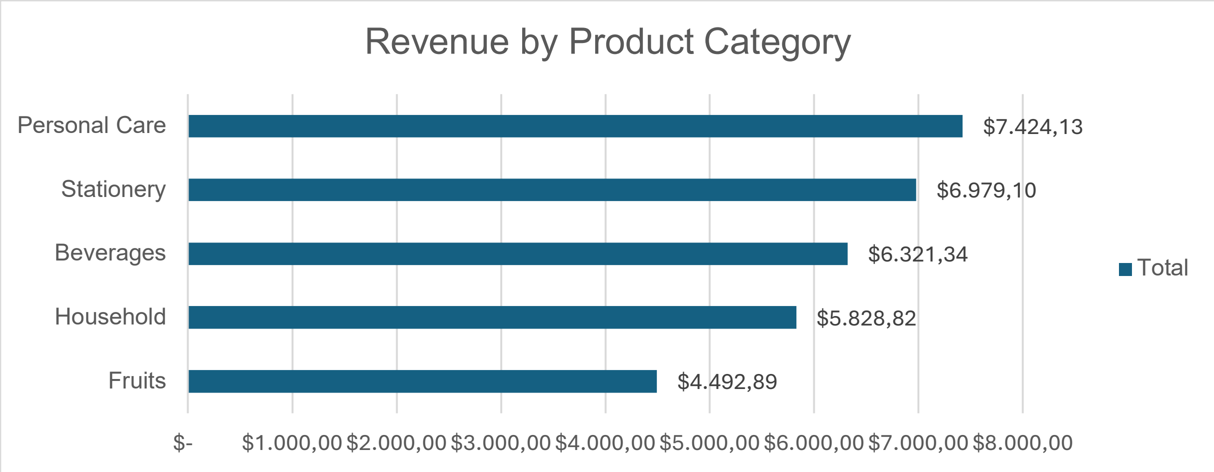
	A	B	C	D	E	F	G	H
1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
21	20	A	Chicago	Normal	Detergent	Household	5	59,75
45	44	A	Chicago	Member	Detergent	Household	4	43,08
62	61	A	New York	Member	Detergent	Household	2	20,09

4. Exploratory Data Analysis & Insights

Insight 1: Revenue Distribution by Product Category

An analysis of total revenue reveals the financial contribution of each product segment. The personal care category is the leading revenue driver ($7,424.13), closely followed by stationery ($6,979.10). This indicates strong consumer demand for essential health and office products. Management should prioritize inventory levels for these categories and consider bundled product packages (e.g., combining stationery with personal care items) to increase the average transaction value.

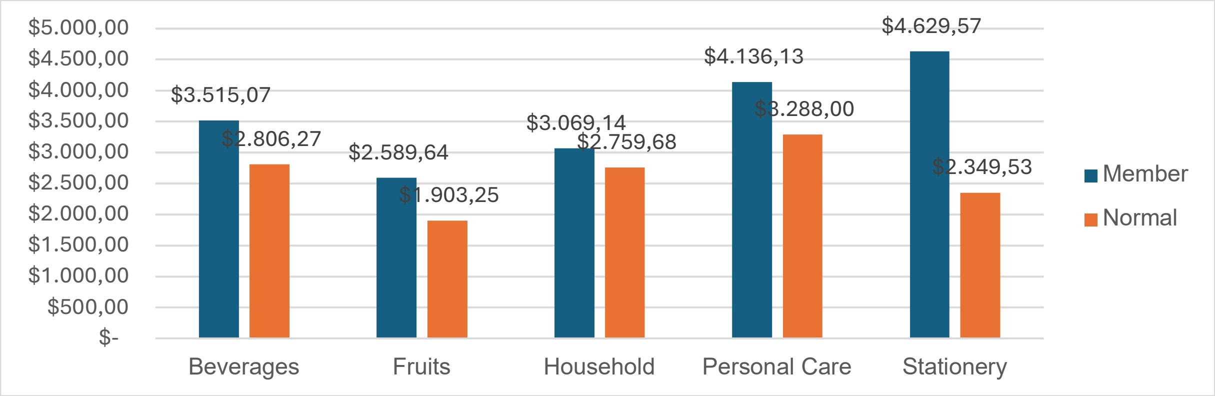
Product Category	Total Revenue
Fruits	$ 4.492,89
Household	$ 5.828,82
Beverages	$ 6.321,34
Stationery	$ 6.979,10
Personal Care	$ 7.424,13
Grand Total	**$ 31.046,28**



Insight 2: Customer Segment Performance & Loyalty Impact

This analysis compares the purchasing power of registered members with that of normal customers (non-registered customers). "Member" customers contribute a significantly higher share of total revenue ($17,939.55) than "Normal" customers ($13,106.73). This gap is most pronounced in the stationery segment. The marketing department should focus on conversion campaigns to encourage regular customers to register as members, as membership demonstrates significantly higher lifetime value for the supermarket.

Total Revenue	Customer Type		
Product Category	**Member**	**Normal**	**Grand Total**
Beverages	$ 3.515,07	$ 2.806,27	$ 6.321,34
Fruits	$ 2.589,64	$ 1.903,25	$ 4.492,89
Household	$ 3.069,14	$ 2.759,68	$ 5.828,82
Personal Care	$ 4.136,13	$ 3.288,00	$ 7.424,13
Stationery	$ 4.629,57	$ 2.349,53	$ 6.979,10
Grand Total	**$ 17.939,55**	**$ 13.106,73**	**$ 31.046,28**


$5.000,00
$4.500,00
$4.000,00
$3.500,00
$3.000,00
$2.500,00
$2.000,00
$1.500,00
$1.000,00
$500,00
$-
$3.515,07
$2.806,27
$2.589,64
$1.903,25
$3.069,14
$2.759,68
$4.136,13
$3.288,00
$4.629,57
$2.349,53
Beverages
Fruits
Household
Personal Care
Stationery
Member
Normal